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Neural connectivity and cortical substrates of cognition in hominoids

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Abstract

Cognitive functions and information processing recruit discrete neural systems in the cortex and white matter. We tested the idea that specific regions in the cerebrum are differentially enlarged in humans and that some of the neural reorganizational events that took place during hominoid evolution were species-specific and independent of changes in absolute brain size.

We used magnetic resonance images of the living brains of 10 human and 17 ape subjects to obtain volumetric estimates of regions of interest. We parcellated the white matter in the frontal and temporal lobes into two sectors, including the white matter immediately underlying the cortex (gyral white matter) and the rest of white matter (core). We outlined the dorsal, mesial, and orbital subdivisions of the frontal lobe and analyzed the relationship between cortex and gyral white matter within each subdivision.

For all regions analyzed, the observed human values are as large as expected, with the exception of the gyral white matter, which is larger than expected in humans. We found that orangutans had a relatively smaller orbital sector than any other great ape species, with no overlap in individual values. We found that the relative size of the dorsal subdivision is larger in chimpanzees than in bonobos, and that the ratio of gyral white matter to cortex stands out in *Pan* in comparison to *Gorilla* and *Pongo*. Individual variability, possible sex differences, and hemispheric asymmetries were present not only in humans, but in apes as well.

Differences in the distribution of neural connectivity and cortical sectors were identified among great ape species that share similar absolute brain sizes. Given that these regions are part of neural systems with distinct functional attributes,

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we suggest that the observed differences may reflect different evolutionary pressures on regulatory mechanisms of complex cognitive functions, including social cognition.

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Introduction

In a series of comparative neuroanatomical morphometric studies of the brains of humans and apes, we showed that human brain evolution and the evolution of complex cognitive capabilities cannot be simply attributed to an overall differential enlargement of the frontal lobe, as both humans and great apes share, relative to the rest of the brain, a large frontal lobe and frontal cortex (Semendeferi and Damasio, 2000; Semendeferi et al., 2002). While it is likely that the absolute size of the human brain is important for human cognition (Gibson, 2002; Passingham, 2002), the outstanding cognitive capabilities of humans may also be supported by discrete modifications in the relative size of specific neural systems that accompanied increases in the absolute size of the brain. In the study reported here, we tested the idea that certain subdivisions of the cerebrum involved in complex cognitive behaviors are differentially enlarged in humans, including sectors of the frontal cortex and subdivisions of the white matter in the frontal and temporal lobes. We also tested the idea that, during hominoid evolution, adaptive neural reorganizational events took place (Holloway, 1968) that are species-specific and possibly independent of changes in absolute brain size (Holloway, 1979, 2001).

We parcellated the white matter into two regions of interest. One was the gyral white matter (GWM), including the sector of the white matter immediately underlying cortical territories; the other region was the core, including all remaining white matter in the frontal and temporal lobes. This type of parcellation of the white matter is analogous to parcellation schemes adopted by other studies, with the gyral white matter corresponding closely to the overlying cortical parcella-

tion unit of the outer zone of the white matter (Makris et al., 1997; Herbert et al., 2004). Furthermore, we identified the three major gross anatomical sectors in the frontal lobe (dorsal, mesial, orbital) and the amount of tissue devoted to gray, as opposed to white, matter (cortex and gyral white matter) within each sector. Enlarged gyral white matter in any region of interest for a given species may indicate increased interconnectivity within and between neighboring cortical regions, many of which are involved in complex cognitive behaviors. Specifically, we tested the hypothesis that humans have relatively more gyral white matter in the frontal and temporal lobes compared to other hominoids. We also tested the idea that orangutans have a smaller orbital frontal cortex than other apes, a hypothesis based on findings from our pilot study, discussed below. Additionally, we analyzed our regions of interest in search of possible asymmetries and sex differences.

Lesions in the frontal lobe were thought, until recently, to result in a generalized frontal lobe syndrome (Hecaen, 1964; Luria and Homskaya, 1964; Teuber, 1964), but systematic studies over the past two decades using the lesion method and imaging technology in humans, as well as a variety of experimental studies in rhesus monkeys, have demonstrated that this large cortical territory comprises anatomical subdivisions with rather distinct functional attributes. The dorsal sector of the frontal lobe is involved in perception, response selection, working memory, and problem solving (Owen, 1997; Bechara et al., 1998; Petrides, 2000; Pochon et al., 2001). It also includes territories activated during language production in humans (Foundas, 2001) and most of the primary motor and premotor cortices (Zilles et al., 1995). The mesial frontal cortex, including the anterior cingulate, is important for processing emotional stimuli

and for production of affective responses (Cummings, 1993; Rezai et al., 1993), planning and initiation of voluntary motor sequences (Tanji and Mushiake, 1996), theory of mind (Fletcher et al., 1995; Gallagher et al., 2000; Stuss et al., 2001), and attention management (Carter et al., 1999; Dagher et al., 1999). The orbital frontal cortex is known to be involved in the evaluation of actions based on emotional reinforcers (Damasio, 1994; Stone et al., 1998; Rolls, 2000).

A number of morphometric studies have addressed the size of individual frontal sectors in the normal (Caviness et al., 1996) and pathological human brain (Andreasen et al., 1994; Mitelman et al., 2003; Carper and Courchesne, 2005). In a pilot comparative study, we measured these three frontal sectors in a small sample of postmortem brains (one of each for the chimpanzee, gorilla, orangutan, and gibbon) and four living humans (Semendeferi et al., 1997). We found the orbital sector of the frontal lobe to be relatively smaller in the orangutan compared to other great apes (Semendeferi et al., 1997). We also found that the three sectors—dorsal, mesial, and orbital—that included both cortex and immediately underlying white matter did not stand out in the human brain.

While complex cognitive functions in humans such as symbolic thought, cognitive planning, decision making, and language production are known to recruit the dorsal, mesial, and orbital frontal sectors (Owen et al., 1996; Bechara et al., 2000; Foundas, 2001; Pochon et al., 2001), the temporal cortex is crucial in the formation and processing of declarative memory (Squire et al., 2004), auditory processing (Poremba et al., 2003), self-recognition (Kircher et al., 2001), visual processing (Mishkin et al., 1983), and the detection of biological motion that underlies theory of mind (Frith and Frith, 1999). Most regions in the temporal lobe have also been implicated in language processing (Damasio et al., 1996; Price, 2000; Gorno-Tempini and Price, 2001; Grabowski et al., 2001; MacSweeney et al., 2002). Nevertheless, none of these cortical territories function in isolation.

Information processing involves the recruitment of discrete neural systems via afferent and efferent axonal connections. The full assortment of connective pathways between cortical regions and

between the cortex and other brain structures forms the white matter. In comparative and evolutionary studies of the brain, this large territory is commonly treated as a uniform entity, despite consisting of numerous and distinct fiber systems, including short and long association fibers (intrahemispheric connections), commissural systems (interhemispheric connections like the corpus callosum), and projection fibers (like corona radiata). The overall spatial arrangement of the fiber systems reflects the topographic relationship between the origins and targets of the connections (Dejerine, 1895; Heimer, 1995). The white matter immediately underlying cortical territories includes, in addition to long projection fibers, short, intrahemispheric, cortico-cortical association fibers that link neighboring cortical regions (Heimer, 1995). It has been suggested that strong connections exist between areas that lie within a given gyrus (on opposite sides), while weaker connections exist between areas on opposing sides of a given sulcus (Van Essen, 1997). Some support for this argument comes from developmental studies of visual cortex in macaques with frontal lobe ablation, where the removal of frontal territories led to formation of convolutions in the visual cortex that would otherwise have been connected to frontal cortex via long distance projections (Goldman-Rakic and Rakic, 1984). In addition, different anatomical subdivisions of the white matter may reflect different maturation events during development. White matter in closer proximity to the cortex myelinates later than core white matter, and myelination occurs late in the first year of life or later, especially in the frontal cortex (Yakovlev and LeCours, 1967). Recent neuroanatomical studies of the human brain have parcellated the white matter using magnetic resonance scans and morphometric techniques similar to those used here to measure the volume of its subdivisions (Makris et al., 1997; Herbert et al., 2004).

Materials

We used scans of the brains of 17 nonhuman primate subjects (from the Yerkes National

Primate Research Center, Emory University) and 10 normal human subjects (from the Department of Neurology, University of Iowa). Nonhuman primate experimental use was approved by the Emory University IACUC, and human subjects provided informed consent in accord with the policies of IRB of the University of Iowa College of Medicine. The human subjects included adult individuals of both sexes (five males and five females with a mean age of 32.4 years, $SD = 8.55$). The nonhuman primates included 14 great apes (five chimpanzees, three bonobos, two gorillas, four orangutans) and three lesser apes (gibbons). The ape subjects included individuals from both sexes; nine individuals were fully adult; the remaining five were subadults. The brains of the subadult individuals were probably at or near adult size, as brain development typically precedes reproductive development (Tanner, 1990). None of the apes in this sample were language-trained. Two of them were wild-born, while the rest were born in captivity and raised either by their mothers or humans (Appendix 1).

Methods

Thin-cut, T1 weighted magnetic resonance images of all human subjects were obtained in a General Electric 4096 Plus Scanner operating at 1.5 Tesla, using the following protocol: SPGR/50, TR 24, TE 5, NEX 2, FOV 24 cm, matrix 256-192, 124 contiguous coronal slices, 1.6 mm thick. The scans of nonhuman subjects were obtained in a Phillips Gyroscan NT; with the exception of one individual for whom the thickness of serial slices was 3.0 mm and slice overlap was 1.5 mm, thickness of the serial axial slices varied between 1.2 and 1.4 mm, with a slice overlap of 0.6 or 0.7 mm (FOV 124-230 cm).

All brains were reconstructed in three dimensions using Brainvox (Damasio and Frank, 1992), an interactive set of programs designed to reconstruct, segment, and measure magnetic resonance images of brains. All segmentations and measurements were calculated using Silicon Graphics workstations. After each brain was segmented from all non-brain tissue and re-

constructed in 3-dimensions (Semendeferi and Damasio, 2000), its contours were traced manually on each coronal slice in both hemispheres. The pons, medulla, cerebellum, and the greater part of the midbrain were excluded using a straight line connecting the two transverse fissures on each coronal slice where these structures were visible. The two hemispheres were separated.

Anatomical regions of interest

For each region of interest, lines were drawn on the surface of each reconstructed hemisphere; these were used as guides for tracing the regions on coronal slices. Each region of interest was defined using major anatomical landmarks present in all species and was traced manually in each coronal section in both hemispheres.

The anatomical boundaries of the frontal and temporal lobes as a whole were identified in a series of previous studies (Semendeferi and Damasio, 2000; Semendeferi et al., 2002). In this study, we removed any portions of the anterior insula that was included in the frontal cortex. Then, the frontal cortex was further subdivided into three sectors: dorsal, mesial, and orbital (Fig. 1). Defining these regions required the identification of landmarks that were consistently present in all species, as described in our pilot study (Semendeferi et al., 1997). Briefly, the frontomarginal sulcus (the principal sulcus in *H. lar*), the lateral orbital sulcus (orbitofrontal sulcus in the apes), and the circular sulcus were used to separate the dorsal sector from the orbital sector of the frontal lobe. The mesial sector was defined as the region extending from the tip of the gyrus rectus to the upper end of the mesial surface (including the mesial part of the superior frontal gyrus), following the edge of the interhemispheric fissure (Fig. 2). We applied gross anatomical and topographical markers for the definition of the three sectors, which is a standard approach in morphometric studies using magnetic resonance images.

The cortex was manually traced on each coronal section in both hemispheres throughout the dorsal, mesial, and orbital frontal sectors and the temporal lobes. In the temporal lobe, the cortex included the hippocampus and the amygdala (Fig. 2).

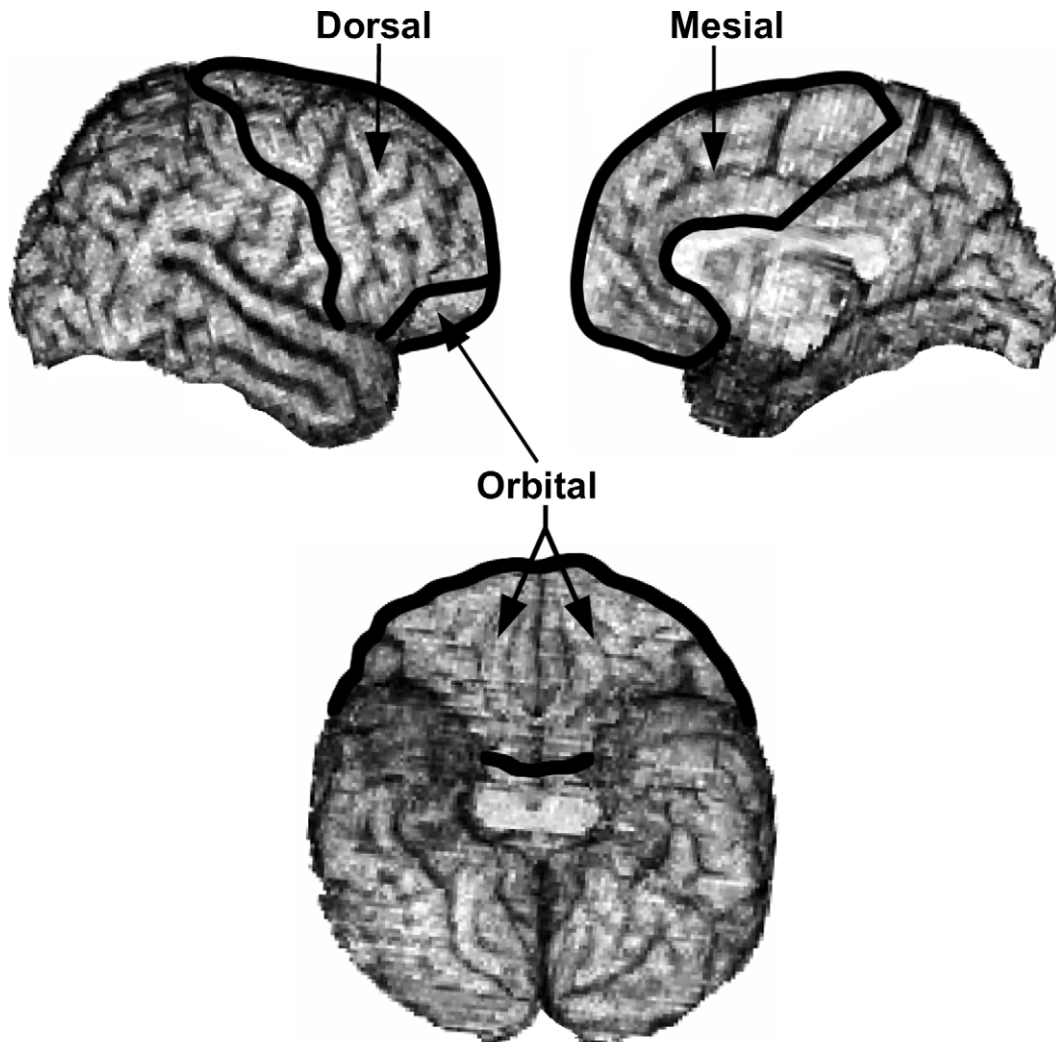


Fig. 1. The dorsal, mesial, and orbital sectors of the frontal lobe in the chimpanzee brain.

A second segmentation separated the gyral white matter from the core white matter and underlying subcortical structures (Fig. 2). This segmentation was achieved by manually drawing straight lines on coronal sections that connected the bottom of each sulcus to the neighboring sulcus at the juncture between the cortex and white matter. All sulci for both hemispheres in each coronal section throughout the regions of interest were traced in this manner.

The resolution of the ape magnetic resonance scans did not allow for an accurate segmentation

of the subcortical structures. Therefore, in the frontal lobe, subcortical structures (e.g., part of the basal ganglia and thalamus) were included in the “core white matter” volume. In the temporal lobe, this region of interest included only white matter (core white matter).

The absolute volume of all regions of interest was calculated from an automatic count of the number of voxels within a region multiplied by the volume of a single voxel in mm^3 . We calculated the total volume of the cortex in the dorsal, mesial, and orbital frontal sectors, the gyral white matter

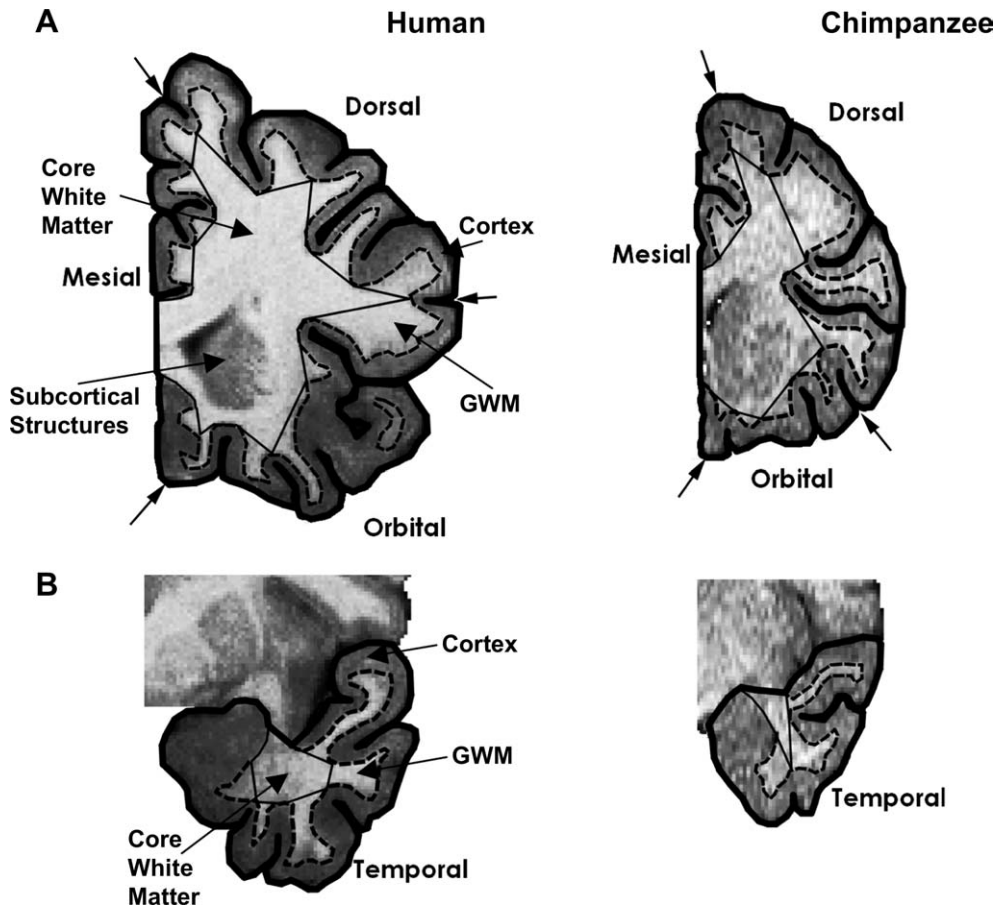


Fig. 2. Coronal sections through (A) frontal and (B) temporal lobes. The boundaries between dorsal, mesial, and orbital frontal sectors are indicated by arrows (A). The dashed line indicates the boundary between white matter and cortex; the thin solid line indicates the boundary between gyral white matter and core white matter; the thick solid line marks the pial surface of the cortex.

underlying each of these cortical sectors, the core sector in the frontal lobe (remaining white matter and subcortical structures), the temporal cortex, the gyral white matter underlying the temporal cortex, and the core white matter in the temporal lobe, as no subcortical structures other than the amygdala are part of this lobe.

A reliability study was undertaken by two tracers comparing the tracings of the cortex—the most arbitrary of the regions of interest—in three randomly chosen subjects (human, orangutan, and gibbon). The cortex was traced in ten coronal sections spaced at equal intervals throughout the right hemisphere. Non-parametric correlations (Spearman's rho) were calculated for each subject

(human, $r = 0.9833$; orangutan, $r = 0.9667$; gibbon, $r = 0.9879$; $p < 0.0001$ for all subjects).

Statistics

We analyzed our data in terms of allometry. Restricted-range log-log regressions were plotted through all species using least squares, except when testing whether human values differed from expectation based on ape data. In those cases, we excluded humans from the regression. We then added the human values to the plot and compared expected versus observed values with respect to the regression line and the prediction interval (95%). A test of proportions, a measure of the proportion

of individual values that fall outside the prediction interval, was applied to evaluate the position of the human values with respect to the prediction interval. We also calculated % residuals for the human values and conducted paired *t*-tests to determine if observed and predicted values were significantly different. In addition to the least squares, we applied the reduced major axis as a best-fit line; we found only minor differences in the relation between observed and expected values between the two approaches.

All regressions were calculated using phylogenetically independent contrasts (PIC) and mapped back into the original data space using the PDTree module of PDAP, a phylogenetic data analysis program (Garland et al., 1999; Garland and Ives, 2000). Independent contrasts were calculated from log-transformed data and standardized using branch lengths derived from Purvis's (1995) composite phylogeny of primates. For all frontal lobe measures, it was necessary to square the branch lengths in order to have standardized contrasts that were not correlated with their standard deviations. The PIC regressions were calculated based on species means, but we plotted all individual values on the graphs. We also calculated regressions using standard data for comparison with the PIC analysis. Furthermore, we calculated the relative size of each region of interest in relation to the volume of other regions of interest. Other statistical tests (paired *t*-tests, Wilcoxon two-sample rank tests) were conducted using JMP IN 4.02 (SAS Institute, 2000).

Results

Dorsal, mesial, and orbital frontal sectors

The dorsal sector was the largest of the three sectors of the frontal cortex, with individual volumes ranging from 29.6 cm³ to 66.5 cm³ in the great apes, 7.2–8.3 cm³ in gibbons, and 137.7–191.5 cm³ in humans. The second largest was the mesial sector, with individual volumes of 12.7–33.1 cm³ in great apes, 3.6–4.2 cm³ in gibbons, and 66.9–90.5 cm³ in humans. The smallest subdivision was the orbital sector, with individual

volumes of 6.9–14.7 cm³ in great apes, 2.1–2.4 cm³ in gibbons, and 30.0–45.6 cm³ in humans (Table 1; Appendix 1).

The dorsal cortex and mesial cortex (Fig. 3A, B) each exhibited isometric relationships with the rest of the frontal cortex (slope = 1.033, SE = 0.094 and slope = 0.988, SE = 0.050, respectively) and no species stood out. Relative volumes of these two regions overlapped among all species (Fig. 4A, B).

The orbital cortex shared a slightly negative allometric relationship with the rest of frontal cortex (slope = 0.902, SE = 0.082; Fig. 3C), although the 95% confidence interval included a slope of 1. Three of the four orangutans fell below the confidence interval, but no other species stood out. The average % residual for the orangutans was –31.99%, meaning that the observed values were significantly smaller than predicted (paired *t*-test, *t*₃ = 9.112, *p* = 0.0028). The orbital cortex occupied a smaller percentage of total frontal cortex in orangutans in comparison to the rest of the species, and the orangutan values were significantly smaller than those of the other great apes (Wilcoxon two-sample rank test, *Z* = 2.758, *p* = 0.0058; Fig. 4C). The only non-orangutan individual that fell within the range of the orangutan values was one of the humans.

Chimpanzees had a relatively larger dorsal sector than did bonobos, regarding both the cortex (not significant; Fig. 4A) and the gyral white matter (Fig. 5; Wilcoxon two-sample rank test, *Z* = 2.087; *p* = 0.037). Orangutans also had a relatively large dorsal cortex, but they did not stand out in terms of gyral white matter in this sector.

The dorsal cortex exhibited a rightward asymmetry across all individuals (Fig. 6A), but this asymmetry only reached significance in the non-human males (paired *t*-test, *t*₁₀ = 2.327, *p* = 0.042). The mesial cortex exhibited a significant leftward asymmetry in nonhuman primates (paired *t*-test, *t*₁₆ = 3.427, *p* = 0.0035; Fig. 6B) that was not present in humans (paired *t*-test, *t*₉ = 0.589, *p* = 0.29). Although there was overlap, the difference between humans and nonhumans was significant (Wilcoxon two-sample rank test, *Z* = 2.536; *p* = 0.011). No asymmetry was apparent

Table 1
Mean volumes for each region of interest¹

Species ²	Dorsal cortex ³	Mesial cortex ³	Orbital cortex ³	Dorsal GWM ⁴	Mesial GWM ⁴	Orbital GWM ⁴	Frontal core ⁵	Temporal cortex	Temporal GWM ⁴	Temporal core ⁶
<i>Homo sapiens</i> (10)	164.2 ± 5.8	75.4 ± 2.9	41.6 ± 1.8	57.7 ± 2.4	18.9 ± 1.3	8.4 ± 0.5	87.1 ± 4.5	163.3 ± 5.9	36.0 ± 1.9	23.8 ± 1.2
<i>Pan troglodytes</i> (5)	36.3 ± 3.6	16.9 ± 1.6	9.6 ± 1.1	12.5 ± 1.3	4.0 ± 0.5	1.6 ± 0.2	18.8 ± 2.3	38.5 ± 3.8	6.3 ± 0.6	5.1 ± 0.8
<i>Pan paniscus</i> (3)	35.7 ± 1.6	18.4 ± 0.3	10.6 ± 1.4	11.9 ± 0.9	4.8 ± 0.6	1.7 ± 0.1	18.3 ± 1.5	39.1 ± 2.2	5.4 ± 0.2	3.6 ± 0.2
<i>Gorilla gorilla</i> (2)	48.0 ± 4.9	23.0 ± 3.3	13.0 ± 1.6	12.7 ± 1.9	5.0 ± 1.0	2.0 ± 0.3	22.1 ± 5.3	40.8 ± 7.1	5.4 ± 0.3	3.4 ± 0.02
<i>Pongo pygmaeus</i> (4)	57.9 ± 4.9	28.5 ± 2.2	9.7 ± 1.1	15.4 ± 1.2	6.2 ± 0.6	1.3 ± 0.2	23.3 ± 2.5	52.5 ± 4.5	7.6 ± 0.7	5.4 ± 0.5
<i>Hylobates lar</i> (3)	7.7 ± 0.3	3.9 ± 0.2	2.3 ± 0.1	2.1 ± 0.2	0.7 ± 0.02	0.2 ± 0.01	2.4 ± 0.1	11.0 ± 0.2	1.2 ± 0.2	0.7 ± 0.2

¹ Values are in cm³ ± S.E. and include both hemispheres.

² Numbers of individuals per species are in parentheses.

³ The dorsal, mesial, and orbital values sum to the total cortex of the frontal lobe.

⁴ GWM = gyral white matter; the dorsal, mesial, and orbital values sum to the total gyral white matter of the frontal lobe.

⁵ Values include core white matter and subcortical structures in the frontal lobe.

⁶ Values include only core white matter in the temporal lobe.

in the orbital cortex, and no significant asymmetries were observed within the gyral white matter in any sector.

Gyral white matter and core white matter

The volume occupied by gyral white matter in the frontal lobe was 11.8–28.3 cm³ in great apes, 2.5–3.3 cm³ in gibbons, and 75.3–114.0 cm³ in humans. The remaining volume of the frontal lobe (12.4–29.4 cm³ in great apes, 2.2–2.6 cm³ in gibbons, and 69.0–111.4 cm³ in humans) not occupied by cortex or gyral white matter was occupied by the core white matter and subcortical structures (mostly basal ganglia). In the temporal lobe, gyral white matter occupied 4.2–9.6 cm³ in great apes, 1.0–1.6 cm³ in gibbons, and 24.6–45.6 cm³ in humans; core white matter (no subcortical structures) occupied 2.5–7.1 cm³ in great apes, 0.5–1.2 cm³ in gibbons, and 17.3–29.4 cm³ in humans, while the remaining volume (28.4–60.0 cm³ in great apes, 10.8–11.4 cm³ in gibbons, and 134.6–190.0 cm³ in humans) was occupied by cortex, including the amygdala and hippocampus (Table 1; Appendix 1).

The relationship between gyral white matter and core in the frontal lobe was isometric when all species were included in the regression (slope = 0.989, SE = 0.0489). Nevertheless, when a regression line was drawn through nonhuman hominoids (Fig. 7A), the best-fit line was negatively allometric (slope = 0.8637, SE = 0.0735). With the exception of one human value that fell below the regression line, all other human values were above the line, with a couple falling outside the confidence interval.

A similar pattern was seen in the temporal lobe, with the relationship between gyral white matter and core white matter being isometric when all species were included (slope = 0.964, SE = 0.0665), but negatively allometric (slope = 0.8292, SE = 0.1050) when only nonhuman hominoids were considered (Fig. 7B). All human values were above the line but within the confidence interval.

There was no significant difference between the regression slopes calculated using phylogenetically independent contrasts and those using standard data (Fig. 7A–D). In the regression lines calculated

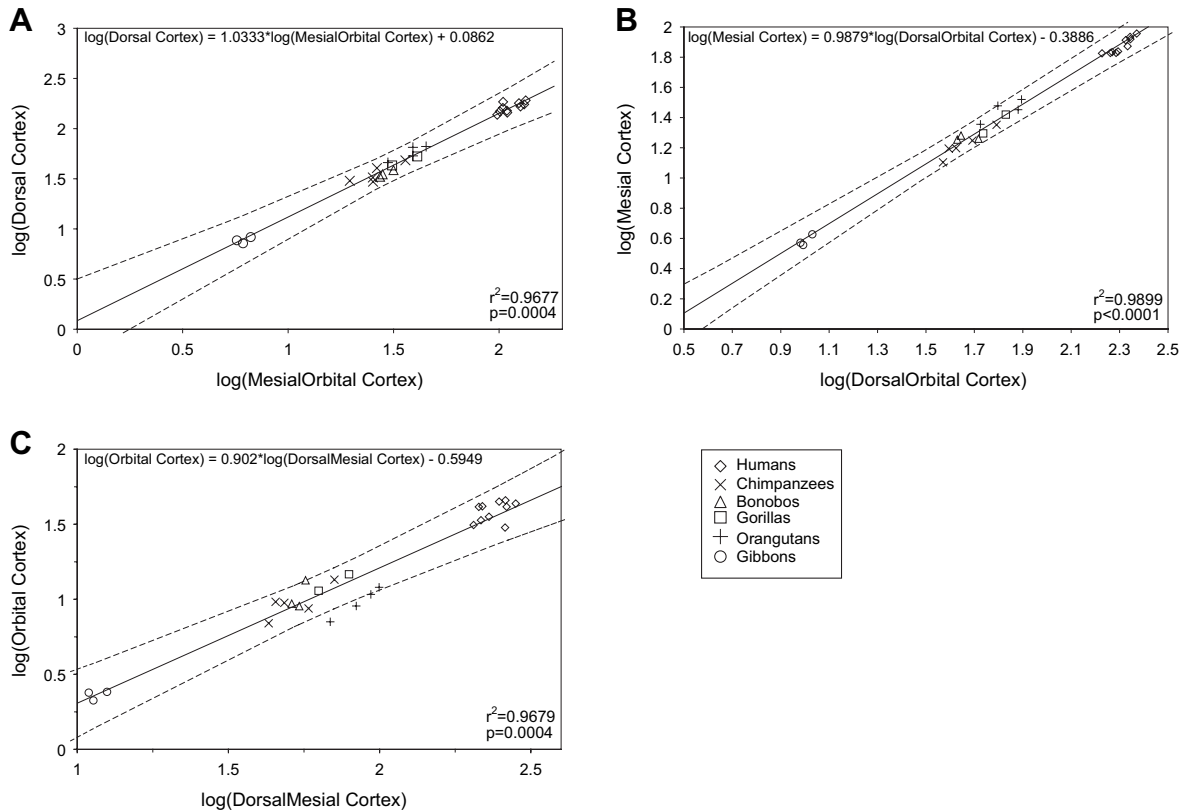


Fig. 3. Log-log regressions through all primates of (A) dorsal cortex, (B) mesial cortex, and (C) orbital cortex. Least squares regressions were calculated using phylogenetically independent contrasts and mapped back into absolute data space.

using nonhuman data only, 6 of 10 human values were outside and above the confidence interval in the frontal lobe, while in the temporal lobe, 5 of 10 human values were above the confidence interval. The average percent residuals of human values for the frontal gyral white matter were 31.5% and 26.3% for the independent contrast and standard regressions, respectively. For the temporal gyral white matter, the average percent residuals of human values were 44.9% and 48.1% for the independent contrast and standard regressions, respectively. Based on residual analysis, the observed human values were significantly greater than predicted in both the frontal and temporal lobes, regardless of the method of regression (paired *t*-test; Frontal: PIC, $t_9 = 6.143$, $p = 0.0002$; standard, $t_9 = 5.152$, $p = 0.0006$; Temporal: PIC, $t_9 = 7.282$, $p < 0.0001$; standard, $t_9 = 7.729$, $p < 0.0001$), and thus the confidence

bands obtained from regressing standard data, which show the humans lying predominantly outside the 95% confidence interval, may be more appropriate for this particular analysis.

The ratio of the temporal core to gyral white matter was larger in females of all species as a group (Wilcoxon two-sample rank test, $Z = 2.097$; $p = 0.036$; Fig. 8). In each species except for gibbons, the average female ratio of temporal core to gyral white matter was larger than the average male ratio. In all species except for humans and orangutans, there was no overlap between individual values for each sex. When gibbons were excluded, we also found that the ratio of total temporal white matter to temporal cortex was larger in females as a group (Wilcoxon two-sample rank test, $Z = 2.196$; $p = 0.028$).

Although differences in absolute size were present between males and females, they were

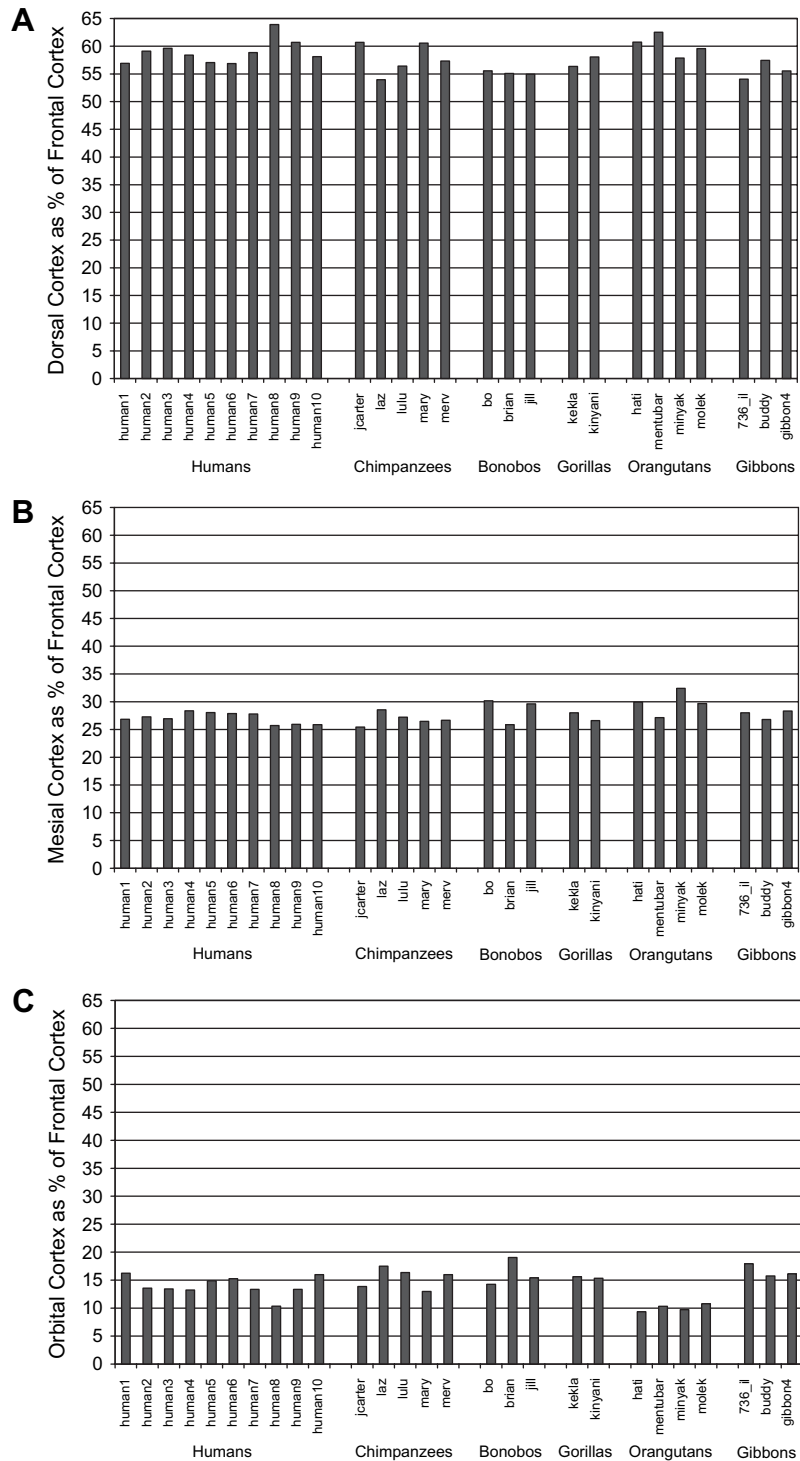


Fig. 4. Relative volumes of (A) dorsal cortex, (B) mesial cortex, and (C) orbital cortex as a percentage of frontal cortex.

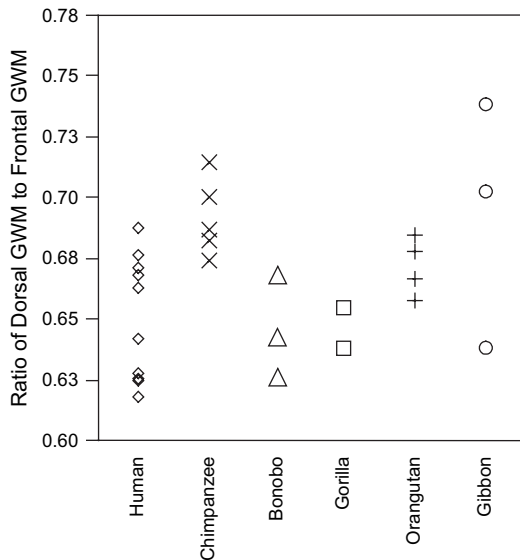


Fig. 5. Ratio of dorsal to frontal gyral white matter (GWM).

not significant and therefore the observed sex differences were unlikely to be related to absolute volume. No sex differences were observed in the ratio of subdivisions of the frontal lobe.

Relationship between white matter and cortex

As expected, we found a positive allometric relationship between total white matter and cortex in both the temporal (slope = 1.266, SE = 0.1155) and frontal (slope = 1.100, SE = 0.0620) lobes. When regressions were drawn through nonhuman primates only, human values were as large as expected. The relationship between the gyral white matter and the cortex in the mesial and orbital frontal sectors was positively allometric (slope = 1.162, SE = 0.0752 and slope = 1.241, SE = 0.0397, respectively). Nevertheless, the dorsal sector had an isometric relationship between gyral white matter and cortex (slope = 1.052, SE = 0.0756). A closer examination of the dorsal sector revealed that chimpanzees and bonobos had an increased ratio of gyral white matter to cortex in comparison to other nonhuman primate species (Fig. 9). Chimpanzee and bonobo values overlapped with human values and had a larger ratio

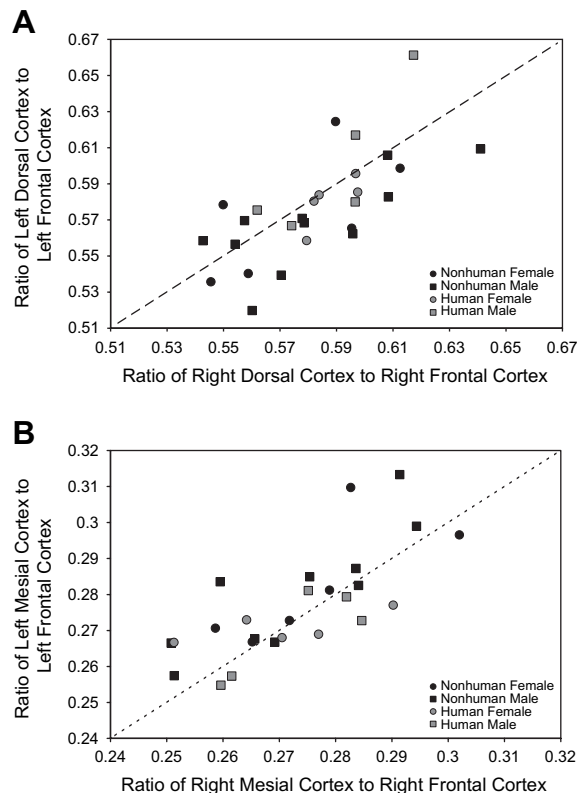


Fig. 6. (A) Relative dorsal cortex asymmetry; (B) relative mesial cortex asymmetry. The dashed line represents perfect symmetry, where values for the ratio are identical in each hemisphere. Points falling above the line reveal a leftward asymmetry, whereas points falling below the line reveal a rightward asymmetry.

of gyral white matter to cortex in the dorsal sector than did gorillas and orangutans (Wilcoxon two-sample rank test, $Z = 2.130$ $p = 0.028$), although there was some overlap in individual values. The gibbon values were similar to those of gorillas and orangutans.

Discussion

Do humans stand out?

Overall, our findings on the size relationships within the three frontal sectors replicated our preliminary report based on a small postmortem

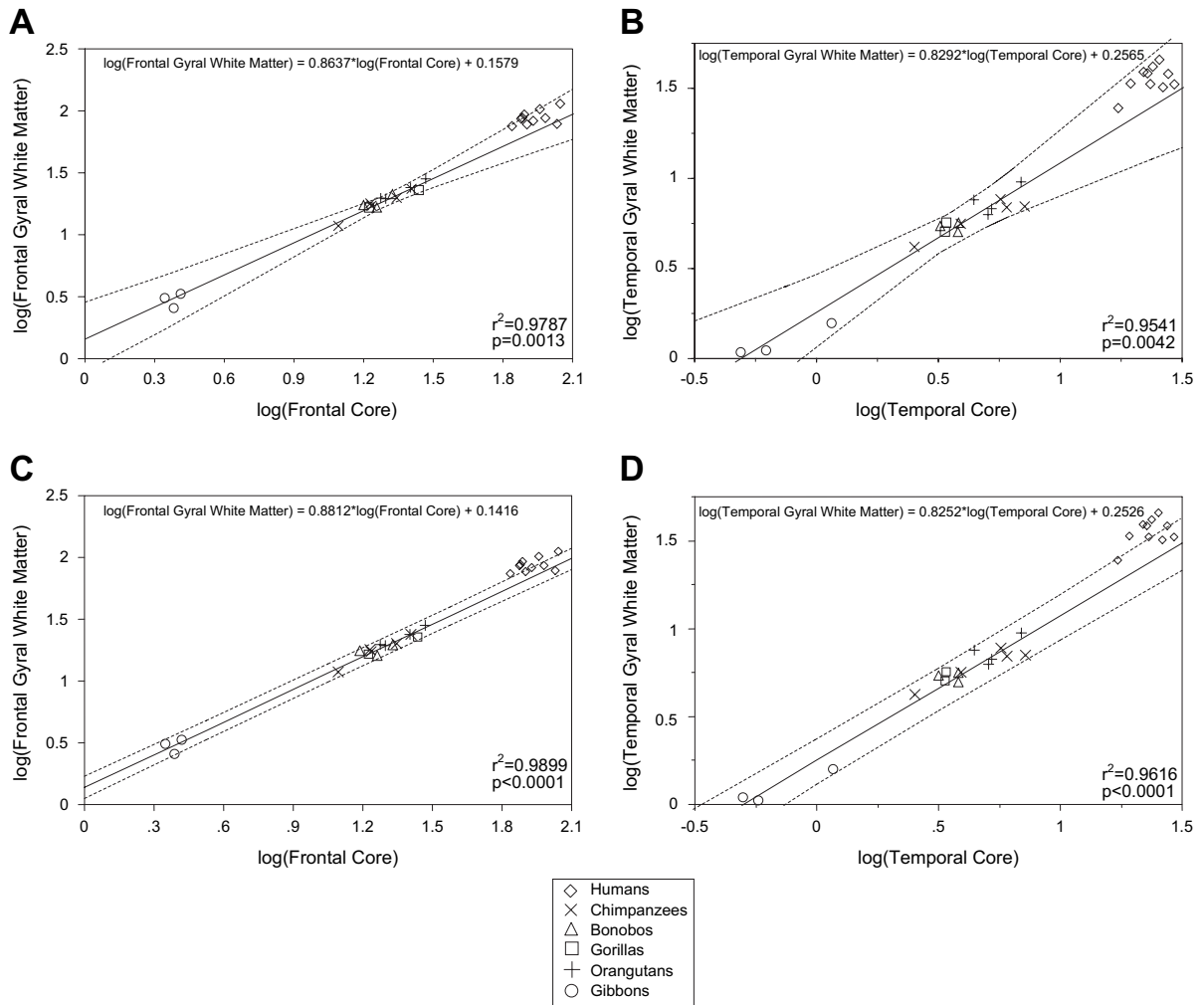


Fig. 7. Log-log graphs of gyral white matter versus core for the frontal lobe and the temporal lobe with least squares regressions calculated using phylogenetically independent contrasts (A, B) and standard data (C, D) through nonhuman primates only. Solid and dashed lines represent the regression lines and their confidence intervals, respectively.

sample (Semendeferi et al., 1997). They were also in agreement with a recent study that used a similar parcellation scheme to analyze these sectors in normal and autistic children and found that the dorsal sector was the largest, followed by the mesial and then the orbital sectors (Carper and Courchesne, 2005). The three sectors maintained their relative size across all individuals, their volumes were highly correlated (Spearman's rho, $r > 0.88$, $p < 0.0001$ for all correlations), and human values did not stand out. We found a positive allometric relationship between the total

white matter and cortex in the temporal and frontal lobes, and the human values were as large as expected. Our finding supports the long established idea that larger brains tend to have a larger percentage of white matter than smaller ones (Frahm et al., 1982; Prothero and Sundsten, 1984; Hofman, 1989; Rilling and Insel, 1999; Zhang and Sejnowski, 2000; Bush and Allman, 2003) and specifically replicates a previous report that used some of the same ape individuals but a different set of human subjects to analyze the primate temporal lobe (Rilling and Seligman,

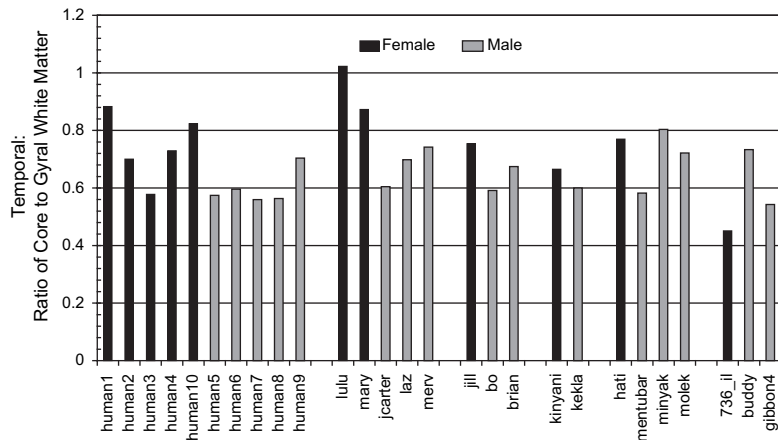


Fig. 8. Ratio of core white matter to gyral white matter in the temporal lobe separated by sex.

2002). However, our findings are at odds with a recent report that suggests that humans have more white matter relative to gray matter than expected (Schoenemann et al., 2005). It is of interest that a re-analysis of the data presented in that study (Sherwood et al., 2005) led to conclusions that agree with the present report.

The scaling relationships between gyral white matter and core white matter in both the frontal and temporal lobes differed from the relationships

in other sectors analyzed. When all species, including humans, were part of the regression, the relationship was isometric and humans did not stand out. Nevertheless, when the regression was based only on the nonhuman hominoids, observed values for humans were larger than expected.

Gyral white matter volume may be related to the degree of cortical gyrification, although direct testing of this idea is needed. The degree of gyrification, measured by a gyrification index, was found to be increased in the prefrontal cortex of humans in comparison to the great apes (Zilles et al., 1988, 1989; Rilling and Insel, 1999), but the temporal cortex was not, with the possible exception of the posterior temporal (Rilling and Insel, 1999). Regardless of whether gyrification proves to be involved in the departure of the human values, there may be functional implications associated with increased gyral white matter volume. White matter is composed predominantly of axons, and axonal width, in addition to axonal numbers, tends to increase with increasing brain size (Harrison et al., 2002). Conduction velocity is faster in wider axons, allowing for more rapid transmission of signals across larger distances. Myelination also increases conduction velocity (Jack et al., 1975, in Harrison et al., 2002), and anatomical asymmetry of white matter in humans has been found to correlate with the thickness of the myelin sheaths that surround axons (Anderson et al., 1999). A differentially enlarged volume of

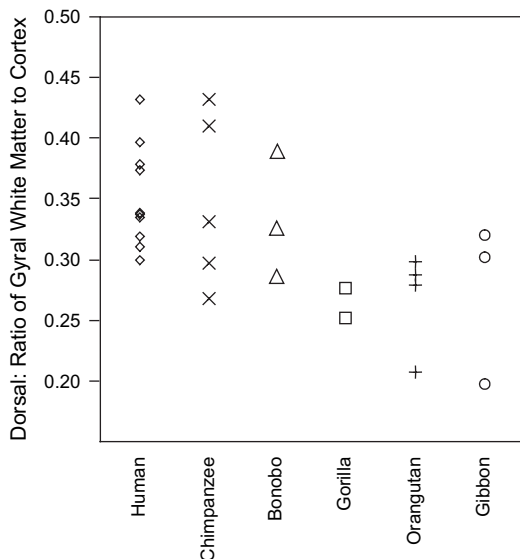


Fig. 9. Ratio of gyral white matter to cortex in the dorsal sector.

gyral white matter, as opposed to core white matter, in humans allows for increased interconnectivity via short association fibers between neighboring cortical areas that, in the case of the frontal and temporal cortex, could be particularly important to human cognitive function. Even if gyral white matter volume was not larger than *expected* in humans, the allometric relationship it shares with the cortex results in humans having an increased percentage of gyral white matter in comparison to the apes (Wilcoxon two-sample rank test; $Z > 2.485$, $p < 0.02$ in all regions analyzed). Differences in the proportion of the brain allocated to a given sector, even if they result from a predictable relationship, cannot be ignored and need to be considered when evaluating their possible functional significance in humans or other species (Passingham, 1982).

Species differences

In spite of having the largest brains among the great apes in our sample, orangutans had some of the smallest values for the orbital sector in terms of absolute size. Orangutans stood out with a relatively smaller orbital sector and no overlap in individual values with any of the other ape species (Fig. 4C). This finding replicates findings from our pilot study (Semendeferi et al., 1997), which included only one orangutan. Extensive studies have demonstrated the connection between the orbitofrontal cortex and aspects of social behavior, including the evaluation of actions based on emotional reinforcers (Bechara et al., 2000; Northoff et al., 2000; Rolls, 2000; Schoenbaum and Setlow, 2001). Patients with orbital lesions in whom these abilities are impaired retain basic cognitive abilities such as memory, attention, and language (Stuss and Benson, 1984; Damasio, 1994). This region is a crucial part of a system that allows individuals to learn social rules and function in close proximity to others on a daily basis.

Unlike other primates, orangutans do not have readily recognizable social groups, and they lead semi-solitary lives. They do congregate into parties regularly, but because of the high costs of feeding competition, party size tends to remain quite small

(Delgado and van Schaik, 2000). As a result, in the wild, orangutans do not encounter many individuals on a daily basis, but it is not clear if this is a relatively recent socioecological phenomenon or a long-standing pattern in the evolutionary history of this taxon. Orangutans raised in captivity, such as the rehabilitants in Borneo, tend to be quite gregarious and even establish close friendships with other individuals (Russon, 2000, 2002). However, even in captive situations, orangutans may differ regarding the complexity of their social interactions in comparison to other primates (Schumacher, pers. comm.).

At least two possible interpretations exist regarding the smaller size of this sector in orangutans. One testable hypothesis is that some cortical areas that are part of the orbital sector in other ape species have shifted toward the lateral surface of the frontal lobe in orangutans, and thus are neither absent nor diminished. The relatively enlarged size of the dorsal cortex in this species (Fig. 4A) supports this idea, but motor-related cortices occupy a substantial portion of that sector and most likely account for the enlargement. Alternatively, the smaller orbital cortex in orangutans reflects different evolutionary pressures with respect to the neural substrates of regulatory mechanisms involved in frequent social interactions. We are in the process of testing the hypothesis that additional components of the neural systems involved in social cognition (e.g., amygdala) are differentially affected in orangutans (Barger et al., 2004).

The size of the dorsal sector as a percentage of the frontal lobe separated the two *Pan* species and was relatively larger in chimpanzees than in bonobos (Fig. 5). The dorsal sector of the frontal lobe includes motor cortices and prefrontal cortical areas involved in perception, response selection, working memory, problem solving, and language.

We hypothesize that differences in motor enervation exist between chimpanzees and bonobos, and that the differences in the relative size of the motor cortices account for the observed enlarged dorsal cortex in chimpanzees. In favor of this hypothesis is the observation that bonobos have much smaller spinal cords than chimpanzees,

despite having similar body size (Rilling and Insel, 1999). It is also of interest that bonobos are not reported to use tools in the wild, whereas chimpanzees are (Tomasello and Call, 1997), and that data on wild populations indicate that the two species differ considerably in their locomotor behavior, with bonobos engaging in more arboreal travel and suspensory behavior than do chimpanzees (Doran and Hunt, 1994). Alternatively, differences may exist in the size of the cortical areas constituting the prefrontal cortex in the dorsal sector. Against this hypothesis is the observation that Brodmann's area 10 in the two species occupies a relatively similar size (Semendeferi et al., 2001). Furthermore, perceived differences in the ability to acquire language via observation, with bonobos displaying greater capacity and efficiency in the acquisition of language than chimpanzees (Savage-Rumbaugh et al., 1992; Rumbaugh et al., 1994), may be less pronounced than once thought (Boysen, 1994).

Within the dorsal region of the frontal lobe, the ratio of gyral white matter to cortex was larger in *Pan* than in either *Gorilla* or *Pongo* (Fig. 9). Primates with larger brains tend to have an increased ratio of white to gray matter, a fact well established in the literature, as discussed above and further supported by our findings. However, chimpanzee and bonobo brains are *not* larger than gorilla and orangutan brains. Furthermore, this relationship was not present in any other frontal or temporal sector examined here. Finally, it did not apply to the core white matter, but only to the gyral white matter component. One possibility, raised above, is that gyral white matter volume, and thus the ratio of gyral white matter to cortex, is related to the degree of cortical gyrification. Although reports exist on cortical gyrification (Zilles et al., 1988; 1989; Rilling and Insel, 1999; Rilling and Seligman, 2002) and cortical folding in primates (Connolly, 1950; Allman, 1998; Falk et al., 2000), these studies do not report individual great ape values, and thus the ability to test this hypothesis awaits future data.

In general, our findings support the argument for conserved evolution and provide support for the idea that the brain “hangs together” (Jerison, 1973;

Finlay et al., 2001). For the most part, the different regions of the brain scale consistently with each other (Spearman's rho, $r > 0.83$, $p < 0.0001$ for all correlations). However, there are some exceptions that support the idea of mosaic evolution (Holloway, 1968; Barton and Harvey, 2000), including the orbital frontal cortex of orangutans, the ratio of gyral white matter to cortex in the dorsal sector of *Pan*, and the relationship between gyral white matter and core white matter in humans. These exceptions suggest that evolutionary forces can also act on individual circuits within the brain without having to act on all, and that such changes can take place independently of overall brain size.

Asymmetries

Here we reported asymmetries in the relative size of the dorsal and mesial subdivisions of the frontal cortex. Specifically, we found a rightward asymmetry in the relative volume of the dorsal frontal cortex (Fig. 6). Although present across all specimens, including humans, the asymmetry reached significance only in the apes. A similar asymmetry in the dorsal frontal lobe has previously been reported in humans (Glicksohn and Myslobodsky, 1993). We also found that the relative volume of the mesial cortex exhibited a leftward asymmetry in the apes in our sample; seven of the ten humans also exhibited a leftward asymmetry, but as a group, the leftward asymmetry did not reach statistical significance in humans. The cingulate gyrus, the anterior portion of which is included in mesial frontal cortex, exhibits a prominent leftward asymmetry in humans (Allen et al., 2002). Although in principle there is no reason to expect that functional asymmetry goes hand in hand with structural asymmetry, it is of interest that functional lateralization has been reported in memory encoding tasks that activate dorsal frontal cortex in humans (Kelley et al., 1998, 2002). Furthermore, it has been suggested that a leftward anatomical asymmetry in the mesial frontal cortex is associated with better performance in both verbal and non-verbal executive tasks (Fornito et al., 2004). The observed leftward asymmetry of the mesial region

may be related to its function in executive control. However, we did not find any asymmetry in the orbital cortex, and there is little evidence to suggest a functional lateralization of orbitofrontal cortex (Nitschke et al., 2004). It is possible that the importance of bilateral function of this region could have stabilized any forces driving asymmetry. No other significant asymmetries were found in the other regions of interest analyzed in this study.

Other studies have also found hemispheric asymmetries in nonhuman primates in specific regions of the frontal (inferior frontal gyrus, Hopkins and Rilling, 2000; Hopkins and Cantalupo, 2003) and temporal lobes (planum temporale, Gannon et al., 1998, 2001). Contrary to conventional wisdom, we saw more asymmetry in these regions in the nonhuman primate brains than in the human ones. Although, this may be due to small sample sizes, the possibility remains that other primate brains are equally or even more asymmetrical in certain sectors than are human brains (Gannon et al., 1998).

Large-scale asymmetries, such as the petalial pattern, with rightward frontal and leftward occipital asymmetry, are well known from the literature (Holloway and De Lacoste-Lareymondie, 1982; Falk et al., 1991), and have been found to be most prominent dorsally in the frontal lobe in humans (Glicksohn and Myslobodsky, 1993). The hemispheres of chimpanzees show comparatively little asymmetry (Gilissen, 2001; but see Hopkins and Marino, 2000). Previously, we identified a rightward asymmetry in the relative size of the frontal lobe as a whole, including white matter and subcortical structures, across all species (Semendeferi, 2001), an observation consistent with other reports (Glicksohn and Myslobodsky, 1993). In the present study, the relative volume of the entire frontal cortex was leftwardly asymmetrical across all individuals studied, in both apes and humans. The frontal core, including subcortical structures, exhibited a rightward asymmetry (paired *t*-test, $t_{26} = 2.329$, $p = 0.028$), while frontal gyral white matter did not. Thus, we hypothesize that the frontal petalia is due to an asymmetry in core white matter or subcortical

structures, such as the basal ganglia and parts of the thalamus. Further testing of this hypothesis awaits studies targeting subcortical structures in the frontal lobe.

Sex differences

Among the great apes, we found that females have more core white matter in the temporal lobe than do males (Fig. 8), but there was no difference between males and females in this ratio in the frontal lobe. Humans also followed this pattern, although there was more overlap between the sexes; gibbons displayed the opposite tendency (males greater than females). These same patterns were seen in the ratio of cortex to total white matter in the temporal lobe. Overall, females appeared to be less asymmetrical than males. Although the present small sample size (five humans for each sex; six and eleven female and male nonhuman primates, respectively) allows only for preliminary conclusions, studies in the human literature also suggest the presence of sex differences in the white matter. Men have more white matter than women in the frontal and parietal lobes or across the cerebrum, although no reports exist on the temporal lobe specifically (McGlone, 1980; Gur et al., 1999; Nopoulos et al., 2000; Allen et al., 2003; Luders et al., 2004).

Attempts to explore a possible relationship between frontal white matter volumes and IQ tests show a correlation to be present in women, while temporal white matter volumes correlate with IQ in men (Haier et al., 2005). McGivern et al. (1998) reported that women display greater unconscious processing of environmental stimuli and may also have better visual recognition memory for familiar objects than do men, an observation that they linked to infant rearing. These functions recruit temporal cortices that are connected via long association fibers to visual cortices in the occipital lobe. Our observation that female primate temporal lobes include an increased amount of core white matter (largely composed of long association fibers) is of interest in the context of McGivern et al.'s (1998) study.

Individual variation

The presence of normal variation in the structure of the human brain has long been appreciated and is supported by recent morphometric studies that report on the size of sectors and lobes in larger samples using techniques similar to those adopted here (Allen et al., 2002; Carper et al., 2002; Carper and Courchesne, 2005). Our results demonstrate the clear presence of individual variation within species, not only in humans, but also in apes, with evidence of more variation within species than between for certain variables. For instance, the mean species values for the relative size of the mesial cortex ranged from 27.1% (human) to 29.7% (orangutan), and the ranges of values overlapped among all species (25.4–32.4%).

We proposed above that, for sectors where ranges of individual variation did not overlap at all (e.g., orbital frontal cortex of orangutans), there might be a species-specific adaptation. But, how does individual variation differ among species, and how is that variation reflected in our sample?

Given the uneven distribution of our sample across species, a reasonable question is whether more variation is present in species represented by larger samples. However, this is not supported by an analysis of coefficients of variation (CV), which reveals no correlation between CV and the number of individuals per species in our sample.

How does variation in humans compare to variation in any given ape species? Is absolute brain size a factor that influences the degree of variation? Do species with larger brains exhibit more variation than those with smaller brains? There was some support for the idea that humans, with their larger brains, are more variable than other primates (e.g., humans versus orangutans in the orbital frontal cortex; Fig. 4C), but generally the CVs for humans fell within the range of CVs found in the other species. Furthermore, the gibbons had larger CVs in certain regions (e.g., gyral white matter) than did species with larger brains.

Another issue is whether variation in the brain is related to the genetic variation within a species. For example, are chimpanzees, which are reported to have a high degree of genetic variation (Gagneux et al., 1999), more variable in brain composition

than others species, especially bonobos and humans? Further studies, with larger sample sizes, are needed to investigate the correlation between genetic variation and variation in the brain.

How is variation influenced by the environment and developmental history of individuals? Neurobiological studies of development support the idea of a plastic nervous system in response to environmental stimuli, especially in the first years of life. Rearing environment has been shown to have profound effects on interhemispheric connectivity, white matter volume, and cognitive ability in rhesus macaques. Infants raised individually in a nursery suffered cognitive deficits that correlate with decreased connectivity and white matter volume, although absolute brain size was not significantly different from controls (Sanchez et al., 1998).

The life histories of individual subjects are relevant to these issues, especially given the constraints on the number of individuals involved in this type of study. Except for two individuals born in the wild (one chimpanzee and one bonobo), the remaining great apes were born in captivity (Appendix 1). Another important factor is that certain great apes in our sample were human-reared (8 of 14), while others were mother-reared (6 of 14). Apes raised by humans in humanlike cultural contexts often acquire new gestures, such as pointing, that are rarely if ever used by individuals that are not exposed to humanlike cultural settings during development (Tomasello and Camaioni, 1997). In addition, captive-born individuals, and in particular those reared by humans, would not have been subjected to the environmental pressures that influence wild-born individuals during development. This provides a basis for testing hypotheses that relate to environmental influences. For example, all of the captive-born chimpanzees in our sample were human-reared, while all of the captive-born bonobos were mother-reared. As discussed above, chimpanzees and bonobos differed in the relative size of the dorsal sector. We do not know what the interplay is, if any at all, between this sector and the effects of the rearing conditions. However, the human-reared individuals may have been exposed to a more diverse environment as juveniles than were the mother-reared individuals, influencing

patterns of dendritic arborization and connectivity early in life.

Our ape sample is small and several observations need greater statistical power. Nevertheless, our neuroanatomical findings support the idea that there is individual variation in neural systems involved in complex cognitive and emotional behaviors, not only in humans, but in apes as well. Goodall (1986) has raised the issue that individual chimpanzees have distinct personalities, and Kanzi, one of the bonobos studied by Savage-Rumbaugh and Lewin (1994), has been portrayed by many as an ape “genius.” Based on the variety of laboratory experiments focusing on ape cognition, it is clear that some individual apes perform better than others in some tasks, much like humans. Ape subjects in studies of cognition also vary in the amount of interest and potential they exhibit in various tasks. Furthermore, interest and potential displayed by an individual can vary with time and social environment (Boysen, 1994). Given the presence of asymmetries, sex differences, and individual variation in the brains of apes, an integration of neuroanatomical findings and individual life histories is important.

Conclusion

Information processing involves the recruitment of discrete neural systems. In this study, we investigated the idea that certain neural systems are differentially enlarged in humans and that some of the ape species have neural specializations that go beyond differences in absolute brain size. Specifically, we addressed the relative composition of white and gray matter within the three major anatomical sectors of the frontal lobe (dorsal, mesial, orbital) and the temporal lobe. We subdivided white matter into gyral white matter and core white matter, as gyral white matter and core white matter are partially involved in different types of information processing. Gyral white matter volume may be related to the degree of interconnectivity among areas that lie in close proximity, whereas core white matter may reflect more long-distance connections.

We found the ratio of gyral white matter to cortex within the dorsal frontal subdivision to be

larger in *Pan* than in either *Gorilla* or *Pongo*. We also found that gyral white matter was relatively larger than expected in humans, not only in relation to cortex, but also in relation to core white matter. This finding may signal differences in how humans and apes process information. An increased ratio of white to gray matter may facilitate fast transmission and possibly faster learning of ever changing stimuli, while an increased ratio of gyral white matter to core white matter may suggest an increased emphasis on short, intrahemispheric, cortico-cortical association fibers. Increased white matter volume via increased numbers of axons, increased cross-sectional area of axons, or increased myelination can produce connections between cortical areas capable of relaying and processing information at greater speeds, facilitating complex cognitive function.

In most regions examined, species with different brain sizes conformed to regular scaling relationships. However, there were some exceptions to the regularity among species that did not differ in absolute brain size. We found that orangutans had a relatively smaller orbital frontal cortex than other great apes, which may reflect different evolutionary pressures with respect to the neural substrates of regulatory mechanisms involved in frequent social interactions.

Our results demonstrate the presence of individual variation, not only in humans, but in other hominoids as well. We identified possible sex differences within our sample; we also found asymmetries in the cortex of the dorsal and mesial sectors, primarily among the apes. These observations need to be replicated with larger samples, including more balanced representations of numbers of individuals per species and analyses incorporating individual life histories.

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Appendix 1

Individual total volumes for each region¹

Species	Individual	Age	Sex ²	Rearing history ³	Frontal cortex	Dorsal cortex ⁵	Mesial cortex ⁵	Orbital cortex ⁵	Dorsal GWM ⁶	Mesial GWM ⁶	Orbital GWM ⁶	Frontal core	Temporal cortex	Temporal GWM ⁶	Temporal core
<i>Homo sapiens</i>	human1	48	F	N/A	254.1	144.6	68.2	41.3	48.5	21.1	8.9	107.9	147.1	33.3	29.4
	human2	22	F	N/A	304.3	179.9	83.0	41.3	54.0	23.5	8.9	75.7	172.2	33.4	23.4
	human3	36	F	N/A	250.0	149.1	67.3	33.6	50.3	18.1	6.9	69.0	163.0	33.6	19.4
	human4	39	F	N/A	235.8	137.7	66.9	31.2	59.6	21.7	6.8	76.5	145.8	38.0	27.7
	human5	31	M	N/A	306.4	174.8	86.0	45.6	66.3	26.6	10.3	91.0	186.9	41.8	24.0
	human6	40	M	N/A	293.1	166.7	81.7	44.7	51.9	22.2	8.8	85.2	171.4	38.3	22.8
	human7	23	M	N/A	325.3	191.5	90.5	43.4	71.6	31.0	11.4	111.4	190.0	45.6	25.5
	human8	24	M	N/A	290.0	185.4	74.6	30.0	62.8	23.7	7.0	78.0	173.9	39.0	22.0
	human9	27	M	N/A	265.5	161.1	68.9	35.5	51.5	18.0	8.2	80.1	134.6	24.6	17.3
	human10	34	F	N/A	260.3	151.3	67.4	41.6	60.1	18.9	8.4	96.2	147.8	32.1	26.4
<i>Pan troglodytes</i>	jcarter	19	M	HR	49.9	30.3	12.7	6.9	8.1	2.8	0.9	12.4	28.4	4.2	2.5
	laz	20	M	HR	54.9	29.6	15.7	9.6	12.8	3.8	1.3	16.9	41.5	5.6	3.9
	lulu	39	F	MR ⁴	57.9	32.6	15.8	9.5	13.4	4.2	2.2	22.0	32.9	7.0	7.1
	mary	36	F	HR	66.9	40.5	17.7	8.7	12.0	3.6	1.6	17.1	38.9	6.9	6.0
	merv	20	M	HR	84.4	48.4	22.5	13.5	16.0	5.7	1.8	25.5	50.8	7.7	5.7
<i>Pan paniscus</i>	bo	25	M	MR ⁴	63.2	35.1	19.1	9.0	13.7	5.8	1.8	21.1	36.2	5.4	3.2
	brian	8	M	MR	70.3	38.7	18.2	13.4	11.1	3.7	1.8	18.1	43.5	5.7	3.8
	jill	11	F	MR	60.6	33.3	17.9	9.3	10.9	4.9	1.6	15.8	37.6	5.0	3.8
<i>Gorilla gorilla</i>	kekla	8	M	MR	93.9	52.9	26.3	14.7	14.6	6.0	2.3	27.4	47.9	5.7	3.4
	kinyani	13	F	HR	74.2	43.1	19.7	11.4	10.8	4.0	1.7	16.7	33.7	5.1	3.4
<i>Pongo pygmaeus</i>	hati	18	F	HR	75.7	46.0	22.7	7.1	13.3	5.4	0.8	19.7	40.2	6.8	5.2
	mentubar	8	M	HR	104.3	65.2	28.3	10.8	13.5	5.0	1.2	18.8	60.0	7.6	4.4
	minyak	15	M	HR	92.8	53.7	30.1	9.0	16.0	6.6	1.4	25.2	51.5	6.3	5.1
	molek	18	M	MR	111.6	66.5	33.1	12.0	18.6	7.8	1.9	29.4	58.2	9.6	6.9
<i>Hylobates lar</i>	736_il	26	F	N/A	13.3	7.2	3.7	2.4	2.2	0.6	0.3	2.2	11.4	1.1	0.5
	buddy	23	M	N/A	13.4	7.7	3.6	2.1	2.5	0.6	0.2	2.6	10.8	1.6	1.2
	gibbon4	5	M	N/A	15.0	8.3	4.2	2.4	1.6	0.7	0.2	2.4	10.8	1.0	0.6

¹ Values are in cm³ and include both hemispheres combined.² M = male; F = female.³ HR = human-reared; MR = mother-reared; N/A = data not available.⁴ All ape individuals are captive-born, except for Lulu and Bo, who were wild-born and most likely mother-reared.⁵ These three values sum to the total cortex of the frontal lobe.⁶ GWM = gyral white matter; these three values sum to the total gyral white matter for the frontal lobe.

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